	National University of Sciences and Technology
	Pakistan Navy Engineering College
	Department of Electronics and Power Engineering
	Quiz-1 (MS Artificial Intelligence – Fall 2026)

Course Code: CS-871 **Course Title:** Machine Learning
Instructors: Dr. Ihsan Ullah Khalil **Timing:** During Class Timing
Max. Marks: 3 **Max. Time: 20** minutes
Name: _____ **Regn No:** _____

S/No.		**Marks**
Q-1	A student fits a 1-Nearest Neighbor ($k = 1$) predictor to a training dataset $(x^1, y^1), \ldots, (x^n, y^n)$, where all feature vectors x^i are distinct. a) What is the predicted value $\hat{y}^i = g(x^i)$ when the model is evaluated on one of its own training points x^i, and what is the resulting training error? b) In machine learning, our ultimate goal is to have $\hat{y} \approx y$ on *unseen* data. Explain conceptually why achieving zero training error with a 1-NN predictor does not guarantee good performance on unseen data.	
Answer		